(ii) Constants of the distributions are:

Mean = np; Variance = npq; Standard deviation = $\sqrt{npq}$

Skewness = $\frac{q-p}{\sqrt{npq}}$; Kurtosis = $\frac{1-6pq}{npq}$

(iii) It may have one or two modes.

(iv) If $X \sim B(n_1, p)$ *and* $Y \sim B(n_2, p)$ and that X and Y are independent then $X + Y \sim B(n_1 + n_2, p)$

(v) If `$n$' independent trials are repeated $N$ times the expected frequency of `x' successes is $N \times {}^nC_x \; p^x \; q^{n-x}$

(vi) If $p = 0.5$, the distribution is symmetric.

Example 10.1

Comment on the following 'The mean of binomial distribution is 5 and its variance is 9'.

Solution:

Given mean $np = 5$ and variance $npq = 9$

$\therefore \frac{Value\ of\ variance}{Value\ of\ mean} = \frac{npq}{np} = \frac{9}{5} \therefore q = \frac{9}{5} > 1$ is not possible

as $0 \leq q \leq 1$ and hence the given statement is wrong.

Example 10.2

Eight coins are tossed simultaneously. Find the probability of getting atleast six heads.

Solution:

Here n=8 $p = P$ (head) $= \frac{1}{2}$ $q = 1 - \frac{1}{2} = \frac{1}{2}$

Trials satisfy conditions of Binomial distribution

Hence $P(X = x) = {}^nC_x \; p^x \; q^{n-x}$ $x = 0, 1, 2, \ldots n$

$= {}^8C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{8-x}$ $x = 0,1,2,\ldots.8$

$= {}^8c_x \left(\frac{1}{2}\right)^{x+8-x}$

$= {}^8C_x (\frac{1}{2})^8$